Canonical Syllogistic A | 4

Aristotle's **canonical syllogistic** is probably the most influential logical system of all times – modern classical logic still has a long wait before getting there.

I will formalise this system representing all inferences admitted by the 192 moods admitted by the 3 Aristotelian figures, among which only 14 are the canonical syllogisms admitted by Aristotle.

4.1 Vocabulary of A.

The vocabulary of the canonical syllogistic can be formalised with: four logical symbols, quantifiers '$\mathtt{A}$', '$\mathtt{E}$', '$\mathtt{I}$', '$\mathtt{O}$'; plus three non-logical ones, the terms 'a', 'b', 'c'. We will not need auxiliary symbols.

Task 4.1. Define (a) '$\mathbf{Log}_{\mathsf{A}}$', (b) '$\mathbf{Non}_{\mathsf{A}}$', and (c) '$\mathbf{Vo}_{\mathsf{A}}$'.

Definition 4.1 (Vocabulary of A (answer to task 4.1))**.**

$$\begin{aligned}
\mathbf{Non}_{\mathsf{A}} &\overset{\text{DF}}{=} \{\text{`}\mathtt{A}\text{'}, \text{`}\mathtt{E}\text{'}, \text{`}\mathtt{I}\text{'}, \text{`}\mathtt{O}\text{'}\}, \\
\mathbf{Log}_{\mathsf{A}} &\overset{\text{DF}}{=} \mathbf{Abc}, \\
\mathbf{Vo}_{\mathsf{A}} &\overset{\text{DF}}{=} \left\langle \mathbf{Log}_{\mathsf{A}}, \mathbf{Non}_{\mathsf{A}} \right\rangle .
\end{aligned}$$

Note that, since $\mathbf{Abc} = \{\text{`}a\text{'}, \text{`}b\text{'}, \text{`}c\text{'}\}$, definition 4.1 entails that $\mathbf{Log}_{\mathsf{A}} = \{\text{`}a\text{'}, \text{`}b\text{'}, \text{`}c\text{'}\}$.

4.2 Formulae of A.

Aristotelian syllogistic deals with *categorical propositions*, which are basic forms of propositions that express a relation between two terms. They are composed of a *subject term*, a *copula*, and a *predicate term*.

Where '$\mathbf{x}$' stands for the predicate term and '$\mathbf{y}$' for the subject term, the basic form of Aristotelian categorical sentences is:

$$\mathbf{x} \text{ is belongs to } \mathbf{y}.$$

For his logical treatment of these expressions, Aristotle considers four kinds of categorical propositions: universal affirmative or *A-type*, universal negative or *E-type*, particular affirmative or *I-type*, and particular negative or *O-type*.

In a formal language, categorical proposition are represented by combining one logical symbol, indicating the kind of proposition,

Table 4.1: Categorical formulae and their natural-language interpretations.

	Affirmative	Negative
Universal	x belongs to all y. $\mathtt{A}xy$	x belongs to no y. $\mathtt{E}xy$
Particular	x belongs to some y. $\mathtt{I}xy$	x does not belong to all y. $\mathtt{O}xy$

and two different non-logical symbols, indicating the terms. Thus, we obtain formulae like '$\mathtt{A}ab$' or '$\mathtt{O}cb$'. Table 4.1 summarises how we represent such sentences..

However, there is no categorical formula such as '$\mathtt{A}bb$', since the system does not permit the same non-logical symbol to occupy both term positions in a formula.

Task 4.2. Define $\mathbf{Fo}_\mathsf{A}$ using $\mathbf{Vo}_\mathsf{A}$.

Hint: Make sure no formula such as '$\mathtt{A}bb$' is included.

Definition 4.2 (Formulae of A (answer to task 4.2))**.**

$$\mathbf{Fo}_\mathsf{A} \overset{\text{DF}}{=} \{❶xy : ❶ \in \mathbf{Log}_\mathsf{A} \mathrel{\wedge\!\!\wedge} x, y \in \mathbf{Non}_\mathsf{A} \mathrel{\wedge\!\!\wedge} x \neq y\}.$$

Note that, by definitions , the following holds:

$$\begin{aligned}\mathbf{Vo}_\mathsf{A} &= \langle \mathbf{Log}_\mathsf{A}, \mathbf{Non}_\mathsf{A} \rangle,\\ \mathbf{Log}_\mathsf{A} &= \{\text{‘A’}, \text{‘E’}, \text{‘I’}, \text{‘O’}\},\\ \mathbf{Non}_\mathsf{A} &= \mathbf{Abc} = \{\text{‘}a\text{’}, \text{‘}b\text{’}, \text{‘}c\text{’}\}.\end{aligned}$$

Hence, the definition above entails that:

$$\mathbf{Fo}_\mathsf{A} = \{❶xy : ❶ \in \{\text{‘A’}, \text{‘E’}, \text{‘I’}, \text{‘O’}\} \mathrel{\wedge\!\!\wedge} x, y \in \{\text{‘a’}, \text{‘b’}, \text{‘c’}\} \mathrel{\wedge\!\!\wedge} x \neq y\}.$$

It is easy to see that $\mathbf{Fo}_\mathsf{A}$ excludes all formulae of the form $\mathtt{Q}tt$, that is, such that it has a repeated term.

Corollary 4.3. For all $❶ \in \mathbf{Log}_\mathsf{A}$ and $t \in \mathbf{Non}_\mathsf{A}$, $❶tt \notin \mathbf{Fo}_\mathsf{A}$.

Proof. Assume, $\mathtt{Q}tt \in \mathbf{Fo}_\mathsf{A}$. However, by definition 4.2, it would follow that $t \neq t$, which is absurd. Thus, $\mathtt{Q}tt \notin \mathbf{Fo}_\mathsf{A}$. □

It is also easy to show that $\mathbf{Fo}_\mathsf{A}$ is finite.

Corollary 4.4. $|\mathbf{Fo}_\mathsf{A}| = 24$.

Proof. We count each instance of the schema which defines $\mathbf{Fo}_\mathsf{A}$, which are the following:

$$\text{‘A}ab\text{’}, \text{‘A}ac\text{’}, \text{‘A}ba\text{’}, \text{‘A}bc\text{’}, \text{‘A}ca\text{’}, \text{‘A}cb\text{’},$$
$$\text{‘E}ab\text{’}, \text{‘E}ac\text{’}, \text{‘E}ba\text{’}, \text{‘E}bc\text{’}, \text{‘E}ca\text{’}, \text{‘E}cb\text{’},$$
$$\text{‘I}ab\text{’}, \text{‘I}ac\text{’}, \text{‘I}ba\text{’}, \text{‘I}bc\text{’}, \text{‘I}ca\text{’}, \text{‘I}cb\text{’},$$
$$\text{‘O}ab\text{’}, \text{‘O}ac\text{’}, \text{‘O}ba\text{’}, \text{‘O}bc\text{’}, \text{‘O}ca\text{’}, \text{‘O}cb\text{’}.$$

These are 24 in total. Hence, $|\mathbf{Fo}_{\mathsf{A}}| = 24$. □

Syllogistic formal languages are very special, since they are the only ones which may be formulated as finite sets of formulae, and with explicit definitions rather than with recursive ones. Neither of these situations will hold for the formal languages of modern logical systems, including the following two.

4.3 Inference relation of A.

Categorical formulae such as ‘Aab’, ‘Eab’, and ‘Obc’ are all well-formed on their own. However, they cannot be combined to form inferences in A (whether valid or not).

Inferences in A must be constructed according to the three *Aristotelian figures**, all of which require exactly two premises and one conclusion. Moreover, the individual term shared by the premises, the *middle term*, should not occur in the conclusion.

*: There is a fourth figure, but which was not explicitly formulated by Aristotle.

Thus, an inference in A has the form:

$$\langle\langle \mathtt{A}, \mathtt{B}\rangle, \{\mathtt{C}\}\rangle,$$

where $\mathtt{A}$ is the major premise, $\mathtt{B}$ is the minor premise, and $\mathtt{C}$ is the conclusion. $\mathtt{A}, \mathtt{B}, \mathtt{C}$ are all categorical formulae of $\mathbf{Fo}_{\mathsf{A}}$.

Thus, where ❶$_1$, ❶$_2$, ❶$_3$ $\in \mathbf{Log}_{\mathsf{A}}$, the three Aristotelian figures may be explained as follows:

- In the *first figure*, the middle term $\mathtt{m}$ occurs as the predicate of the major premise and as the subject of the minor premise:

  ❶$_1\mathtt{xm}$, ❶$_2\mathtt{my}$ / ❶$_3\mathtt{xy}$.

- In the **second figure**, the middle term $\mathtt{m}$ occurs as the predicate in both premises:

  ❶$_1\mathtt{mx}$, ❶$_2\mathtt{my}$ / ❶$_3\mathtt{xy}$.

- Finally, in the **third figure**, the middle term $\mathtt{m}$ occurs as the subject in both premises:

  ❶$_1\mathtt{xm}$, ❶$_2\mathtt{ym}$ / ❶$_3\mathtt{xy}$.

In order to formalise, we will construct inferences so that '*c*' always stands for the middle term, '*a*' for the subject term of the conclusion and '*b*' for the predicate term of the conclusion.

Task 4.3. Define '$\mathbf{In}_{\mathsf{A}}$'.

Definition 4.5 (Inference in A (answer to task 4.3)).

$$\begin{aligned}\mathbf{In}_{\mathsf{A}} \overset{\mathrm{DF}}{=} &\{\text{🕓}_1 ac, \text{🕓}_2 cb \ / \ \text{🕓}_3 ab : \text{🕓}_1, \text{🕓}_2, \text{🕓}_3 \in \mathbf{Log}_{\mathsf{A}}\} \cup \\ &\{\text{🕓}_1 ca, \text{🕓}_2 cb \ / \ \text{🕓}_3 ab : \text{🕓}_1, \text{🕓}_2, \text{🕓}_3 \in \mathbf{Log}_{\mathsf{A}}\} \cup \\ &\{\text{🕓}_1 ac, \text{🕓}_2 bc \ / \ \text{🕓}_3 ab : \text{🕓}_1, \text{🕓}_2, \text{🕓}_3 \in \mathbf{Log}_{\mathsf{A}}\}.\end{aligned}$$

As with $\mathbf{Fo}_{\mathsf{A}}$ the cardinality of $\mathbf{In}_{\mathsf{A}}$ is finite and can be easily established.

Corollary 4.6. $|\mathbf{In}_{\mathsf{A}}| = 192$.

Proof. Easily (though tediously) shown by counting the instances. Alternatively, to each figure corresponds $4 \times 4 \times 4 = 64$ inferences. Since there three such figures, we have $64 \times 3 = 192$. □

4.4 Consequence relation of A.

Out of the 192 inferential combinations included in $\mathbf{In}_{\mathsf{A}}$, Aristotle accepts exactly 14 as syllogisms – in our terms, as A-valid inferences.

It must be noted that the sense of a categorical expression is similar to the sense in which it would be interpreted by modern standard logic. The main important difference is that, in expressions of the forms $\ulcorner \mathsf{A}\mathtt{xy} \urcorner$ (x belongs to all y) and $\ulcorner \mathsf{I}\mathtt{xy} \urcorner$ (i.e., x belongs to no y), we assume that there exists at least one y.

Definition 4.7 (Consequence in A).

$$\begin{aligned}\mathbf{Con}_{\mathsf{A}} \overset{\mathrm{DF}}{=} \{&\langle\langle \mathsf{A}ac, \mathsf{A}cb\rangle, \{\mathsf{A}ab\}\rangle, \langle\langle \mathsf{E}ac, \mathsf{A}cb\rangle, \{\mathsf{E}ab\}\rangle, \\ &\langle\langle \mathsf{A}ac, \mathsf{I}cb\rangle, \{\mathsf{I}ab\}\rangle, \langle\langle \mathsf{E}ac, \mathsf{I}cb\rangle, \{\mathsf{O}ab\}\rangle, \\ &\langle\langle \mathsf{E}ca, \mathsf{A}cb\rangle, \{\mathsf{E}ab\}\rangle, \langle\langle \mathsf{A}ca, \mathsf{E}cb\rangle, \{\mathsf{E}ab\}\rangle, \\ &\langle\langle \mathsf{E}ca, \mathsf{I}cb\rangle, \{\mathsf{O}ab\}\rangle, \langle\langle \mathsf{A}ca, \mathsf{O}cb\rangle, \{\mathsf{O}ab\}\rangle, \\ &\langle\langle \mathsf{A}ac, \mathsf{A}bc\rangle, \{\mathsf{I}ab\}\rangle, \langle\langle \mathsf{E}ac, \mathsf{A}bc\rangle, \{\mathsf{O}ab\}\rangle, \\ &\langle\langle \mathsf{I}ac, \mathsf{A}bc\rangle, \{\mathsf{I}ab\}\rangle, \langle\langle \mathsf{A}ac, \mathsf{I}bc\rangle, \{\mathsf{I}ab\}\rangle, \\ &\langle\langle \mathsf{O}ac, \mathsf{A}bc\rangle, \{\mathsf{O}ab\}\rangle, \langle\langle \mathsf{E}ac, \mathsf{I}bc\rangle, \{\mathsf{O}ab\}\rangle\}.\end{aligned}$$

4.5 Interpretation and derivation systems for A

Since we have a complete definition of $\mathbf{Con}_{\mathsf{A}}$, it may seem useless to construct an interpretation or a derivation system for A.

If the aim is just to formalise A, it may be useless. However, if we want to extend A into a system admitting inferences of more than two premises. Although we do not know that Aristotle had develop systematically a syllogistic more than two premises, this development is consistent with his interests. In fact, he makes some syllogistic chains in his works, articulating various syllogisms to produce a conclusion from more than to premises.

We will not develop such semantics and calculus, though.